

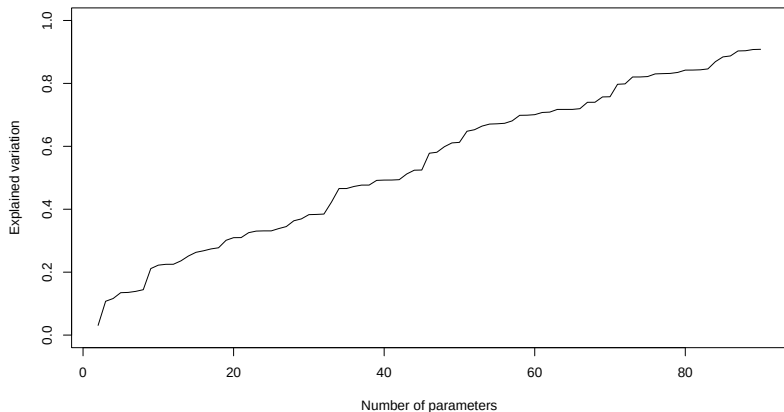
Variance explained and partitioning

Bert van der Veen

University of Bayreuth

The problem of model complexity

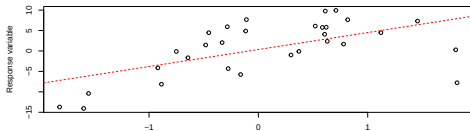
A model with always fit better if you add a parameter



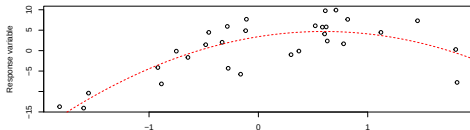
Model complexity



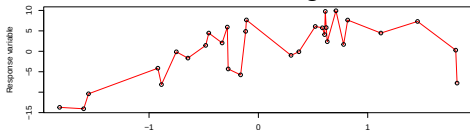
Underfitting



Just right

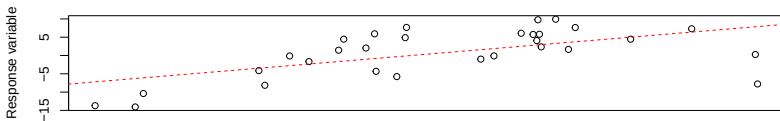


Overfitting

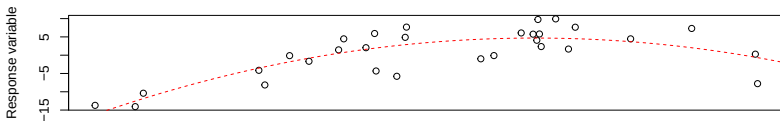


Explained variation

Low R^2



Intermediate R^2



High R^2

R^2 : coefficient of determination

The R^2 statistic helps us to assess how much variability we have explained.

Low R^2 : not so much?

High R^2 : too much?

- ▶ Usually larger than 0 but smaller than 1
- ▶ Observational studies usually have low R^2
- ▶ Experimental studies usually have high R^2

R^2 **cannot be used to assess goodness of fit**

R^2 in linear regression

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \mu_i)^2}{\sum_{i=1}^n (y_i - \alpha_i)^2} \quad (1)$$

- ▶ the numerator represents the unexplained variation of a model
- ▶ the denominator represents the explainable variation of a model
- ▶ One minus their fraction is the explained variation

Adjusted R^2

Similar to information criteria, we can add a penalty for complexity

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$$R^2_{adjusted} = 1 - (1 - R^2) \frac{(n - 1)}{n - p - 1} \quad (2)$$

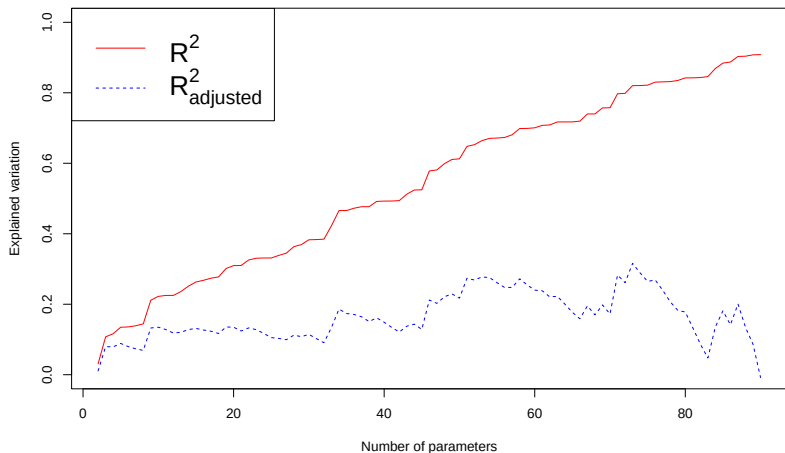
Adjusted R^2

Similar to information criteria, we can add a penalty for complexity

$$R^2_{adjusted} = 1 - (1 - R^2) \frac{(n - 1)}{n - p - 1} \quad (2)$$

- ▶ Negative with $p > n$, otherwise always lower than R^2
- ▶ Penalize the statistic for increased complexity
- ▶ Penalty is large when p approaches n
- ▶ Such as for small samples

Visually: R^2 and adjusted



partial R^2

How much variation is explained by a covariate?

$$R_k^2 = 1 - \frac{\sum_{i=1}^n (y_i - \mu_i)^2}{\sum_{i=1}^n (y_i - \alpha - x_{ik}\beta_k)^2} \quad (3)$$

Allows us to partition the explained variation over the covariates.

- ▶ Quantifies the variation additionally explained in a more complex models
- ▶ Is sensitive to order of covariates
- ▶ Multicollinearity: two covariates can explain similar variation

partial R^2

How much variation is explained by a covariate?

$$R_k^2 = 1 - \frac{\sum_{i=1}^n (y_i - \mu_i)^2}{\sum_{i=1}^n (y_i - \alpha - x_{ik}\beta_k)^2} \quad (3)$$

- ▶ the numerator represents the unexplained variation of a model
- ▶ the denominator represents the explainable variation of a **covariate**
- ▶ One minus their fraction is the explained variation

Connection to Variance Inflation Factor (VIF)

$$\text{VIF}_k = \frac{1}{1 - R_k^2} \quad (4)$$

Measures increase in uncertainty of a parameter estimator when changing the model.

Pseudo R^2 in Generalised Linear Models

- 1) GLMs lack the error term of LMs
- 2) There is often no (clear) residual variance parameter
- 3) R^2 is thus not clearly defined
- 4) Many many different R^2 statistics exist

Concepts and name remain the same.

GLM R^2

R^2 in linear models is based on the error

- ▶ GLMs do not have an explicit error term
- ▶ Defining an R^2 thus not as intuitive
- ▶ A similar interpretation not possible

But, we can still use it as an explorative statistic

A few GLM R^2 's

- ▶ Deviance R^2
- ▶ Pseudo R^2
- ▶ McFadden's R^2
- ▶ Cohen's R^2
- ▶ Tjur's R^2
- ▶ Somer's R^2
- ▶ Cox and snell R^2
- ▶ Nagelkerke R^2
- ▶ Efron's R^2
- ▶ Many more (see `performance::R2`)
- ▶ which one do we choose?

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Remember: **deviance** is the GLM equivalent of RSS!

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Deviance R^2

Remember: **deviance** is the GLM equivalent of RSS!

$$\text{deviance } R^2 = 1 - \frac{D(\mathbf{y}; \hat{\boldsymbol{\mu}})}{D(\mathbf{y}; \hat{\boldsymbol{\alpha}})}$$

$$\text{Cohen } R^2 = \frac{D(\mathbf{y}; \hat{\boldsymbol{\alpha}}) - D(\mathbf{y}; \hat{\boldsymbol{\mu}})}{D(\mathbf{y}; \hat{\boldsymbol{\alpha}})}$$

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Remember: **deviance** is the GLM equivalent of RSS!

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$$\text{deviance } R^2_{adjusted} = 1 - \left\{ 1 - \frac{D(\mathbf{y}; \hat{\boldsymbol{\mu}})}{D(\mathbf{y}; \hat{\boldsymbol{\alpha}})} \right\} \frac{n-1}{n-p}$$

Based on reduction in deviance

McFadden R^2

$$\text{McFadden } R^2 = 1 - \frac{\log \mathcal{L}(\mathbf{y}; \Theta_1)}{\log \mathcal{L}(\mathbf{y}; \Theta_0)} \quad (5)$$

Based on reduction in the ratio of log-likelihoods.

GLM R^2 's

All quantify change in fit due to increased model complexity, in one way or another.

Which one do we use?

Example: Lizards interaction

First model without interaction of Time and Site

```
model1 <- glm(cbind(ghrami, opalinus)~Time+Site,
              data = lizards, family="binomial")
```

Second model with interaction

```
model2 <- glm(cbind(ghrami, opalinus)~Time*Site,
              data = lizards, family="binomial")
```

Lizards: R^2 under more complex model

```
nullmodel <- update(model1, formula = .~1)
(devianceR2 <- c(1-deviance(model1)/deviance(nullmodel), 1-deviance(model2)
```

```
## [1] 0.2168604 0.2325368
```

```
(adjdevianceR2 <- 1-(1-devianceR2)*(n-1)/(n-c(attr(logLik(model1), "df"), a
```

```
## [1] 0.093206835 0.006812369
```

```
(mcfaddenR2 <- c(1-logLik(model1)/logLik(nullmodel), 1-logLik(model2)/logL
```

```
## [1] 0.1197730 0.1284312
```


Variability of R^2

R^2 is a statistic of the data

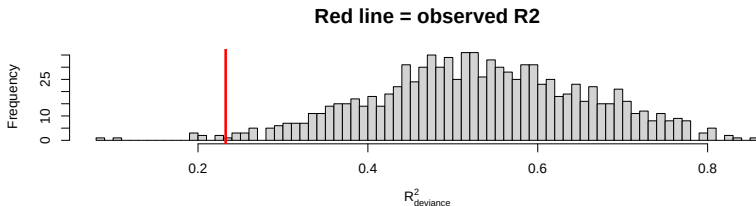
So it is susceptible to sampling variation! (such like LRT and information criteria)

Veall and Zimmermann (1992)

- ▶ Evaluate which R^2 best mimics the original statistic
- ▶ Conclude that various popular R^2 severely underestimate the truth

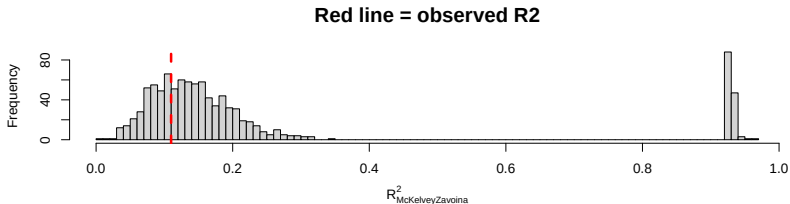
Lizards: deviance R^2

```
R2s <- NULL
for(i in 1:1000){
  set.seed(i)
  ynew <- as.matrix(stats::simulate(model2))
  model2s <- update(model2, formula = ynew~.)
  nullmodels <- update(model2s, formula = .~1)
  R2s <- c(R2s, 1-deviance(model2s)/deviance(nullmodels))}
```



Lizards: McKelvey-Zavoina R^2

```
R2s2 <- NULL
for(i in 1:1000){
  set.seed(i)
  ynew <- as.matrix(stats::simulate(model2))
  model2s <- update(model2, formula = ynew~.)
  nullmodels <- update(model2s, formula = .~1)
  R2s2 <- c(R2s2, DescTools::PseudoR2(model2s, "McKelveyZavoina"))
}
```



Conclusion

- ▶ R^2 to quantify improvement of fit in GLMs
- ▶ There is no single agreed upon statistic
- ▶ Careful with overinterpretation: low R^2 can be just fine